Individual Assignment Cover Sheet

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Introduction

This work summarises an assessment towards how retail mutual funds managers can evaluate the impact of investing in polluting and not polluting US stock firms. In consistency with other courses of the MBA, it is clear the financial sector has increasingly recognised the importance of sustainability, with pollution emerging as a key consideration in investment decisions. The United Nations Sustainable Development Goals (SDGs**)**, particularly Goal 13 (Climate Action) and Goal 12 (Responsible Consumption and Production), emphasise the need for businesses to reduce their environmental impact (United Nations, 2024). As it is the cases in UK, USA and Europe, their governments are introducing stricter regulations and investors shift towards Environmental**,** Social, and Governance (ESG) investing. As it is stated along this assessment, pollution can be considered both a risk factor and a behavioural factor affecting stock performance.

For retail mutual funds as per the simulated PMC scenario of the assignment, this dual perspective is crucial. In general, high-pollution (P) firms may offer higher returns but face greater regulatory and transition risks, aligning with SDG-driven policies that discourage unsustainable practices. In contrast, low-pollution firms (C) may benefit from the increasing preference for ESG investments but risk overvaluation due to excessive demand. Understanding how pollution influences asset pricing can help retail mutual funds better deal with these complexities while contributing to global sustainability efforts. By the end of the discussion, a formal statement is presented on whether pollution is found to be a risk or a behavioural factor.

Discussion

Regulations and rankings

Over more recent years, sustainable investing seems to align with global initiatives like the United Nations Sustainable Development Goals (SDGs). This is mean to address concerns about the cost of sustainability. According to leading financial publication firms that prioritise sustainability often face higher operational costs, such as investing in cleaner technologies, complying with stricter regulations, and meeting ESG reporting requirements (The Economist, 2018).

The Corporate Knights ranking provides investors with a transparent evaluation of the world's most sustainable companies, helping them align investments with Environmental, Social, and Governance (ESG) criteria. By analysing key sustainability metrics such as carbon emissions, resource efficiency, and ethical governance, the ranking aids in identifying firms with long-term growth potential and lower regulatory risks (Corporate Knights, 2024). This supports informed, responsible investment decisions.

US Stocks and pollution scrutiny

Stocks of US firms are increasingly subject to higher levels of pollution scrutiny, particularly from mutual funds that integrate environmental, social, and governance (ESG) criteria into their investment strategies. Regulatory bodies, such as the Securities and Exchange Commission (SEC), have introduced enhanced climate-related disclosure requirements, compelling publicly traded companies to report their carbon emissions and sustainability risks (SEC, 2022, pp. 15-18). Additionally, institutional investors and mutual funds are aligning their portfolios with sustainability goals, applying stricter screening measures to exclude or underweight high-pollution firms. This shift is driven by the growing financial risks associated with environmental non-compliance, including potential litigation, carbon taxation, and reputational damage, which can negatively impact stock valuations and investor confidence.

Furthermore, the rise of sustainable investment indices and corporate ranking systems, such as the Corporate Knights Global 100, previously introduced, has intensified scrutiny on US firms' pollution levels. Mutual funds managing ESG-focused portfolios must consider how firms comply with evolving environmental regulations and consumer expectations regarding sustainability (Corporate Knights, 2024). Firms with weak pollution controls face divestment risks as ESG-driven capital shifts towards lower-carbon companies. This increased scrutiny affects asset pricing and sectoral capital flows, reinforcing the need for mutual funds to incorporate pollution risk assessments into their investment strategies.

Mutual funds, risks and behavioural factors

Mutual funds collect capital from multiple investors to create a diversified portfolio of assets, managed by professional fund managers. Diversification helps reduce unsystematic risk while asset allocation strategies manage exposure to broader market risks (Reilly and Brown, 2021). To handle risk factors, such as economic downturns or regulatory changes, mutual funds use portfolio diversification, hedging techniques, and sector rotation (Elton et al., 2020). Behavioural factors, such as investor biases and market sentiment, are addressed by mutual funds through investment strategies, contrarian approaches, and ESG integration to avoid mispricing and overvaluation. By balancing these factors, mutual funds aim to achieve consistent, risk-adjusted returns (Barberis and Thales, 2003). These techniques can be relevant for the current focus on whether it is possible to optimise the performance of the investment portfolio presented for the polluting and non-polluting firms.

Applicability on retail mutual funds

For a retail mutual fund, the scenario presented for Clean vs Polluting firms, creates a trade-off between ethical investing and financial returns. High-pollution firms may offer higher expected returns as compensation for regulatory and transition risks, while low-pollution firms may be benefited by a green premium, leading to overvaluation and

lower future returns. The mutual fund must balance sustainability goals with financial performance, with consideration on assessing whether the cost of sustainability really leads to potential long-term benefits. This makes clear the need for strategic asset allocation integrating risk-adjusted returns and ESG considerations, ensuring long-term profitability without compromising environmental responsibility.

Pollution related factors

Pollution is considered as a risk factor when it is linked to risk exposure, as investors demand higher returns for holding stocks of firms exposed to regulatory costs, climate risks, and transition risks (Engle et al., 2020, pp. 45-48). Stricter environmental policies and carbon pricing increase operational costs for high-emission firms, affecting profitability and market valuation (Ilhan, Sautner and Vilkov, 2021, pp. 320-323). These financial risks contribute to the higher expected returns of polluting firms. This in turn reflects the market's compensation for heightened uncertainty.

This assessment aligns with identifying pollution mainly as a risk factor rather than behavioural factor. This is based on the previous statements on the regulatory ecosystem initiated by the UN Sustainable Development Goals, Corporate Knights Rankings and the higher returns expected by investors. A key approach to mitigating this risk factor in retail mutual fund strategies is the use of an established factor model in investment decision-making. By incorporating recognised factor modelling techniques, mutual funds can better assess and compensate for risk exposure, enhancing portfolio diversification and return optimisation (Fama, E. F. and French, 2015).

Fama-French factor modelling

Incorporating environmental factor modelling into investment strategies is essential for mutual funds aiming to achieve long-term, risk-adjusted returns. Sustainability concerns are increasingly influencing asset pricing, making it crucial for funds to integrate ESG considerations into their portfolio construction. A key approach involves tilting investments towards low-carbon firms while maintaining sufficient diversification to avoid overvaluation. Factor-based models that account for environmental risks allow mutual funds to balance return potential with sustainability objectives (Ilmanen, 2022). By integrating environmental factor modelling, the retail mutual fund can effectively navigate market uncertainties while aligning with long-term sustainability trends.

The Fama-French three-factor model is relevant for risk factors in mutual fund investments, as it explains asset returns based on market risk, size, and value factors (Fama and French, 1993, pp. 5-8). This approach assumes that investors require compensation for systematic risks rather than being driven by behavioural biases. While behavioural factors can influence short-term pricing inefficiencies. This model aligns with risk-based explanations of stock returns, guiding mutual funds in constructing diversified portfolios to optimise risk-adjusted performance (Bodie, Kane and Marcus, 2021, pp. 334-338).

Conclusion

This assessment has explored the dual perspective of pollution as both a risk factor and a behavioural factor in the context of retail mutual fund investments. Through an analysis of regulatory frameworks, sustainability rankings, and financial modelling techniques, it has been demonstrated that pollution aligns more closely with a risk-based perspective, specially around stocks of US firms. Stricter environmental policies, carbon pricing, and transition risks create financial uncertainties for polluting firms, necessitating higher expected returns to compensate for these risks. The Corporate Knights ranking and the UN Sustainable Development Goals reinforce the increasing importance of sustainability in investment decisions, further shaping market expectations and asset pricing.

For a retail mutual fund, balancing financial performance with sustainability considerations remains a key challenge. The trade-off between investing in high-pollution (P) firms with higher returns and low-pollution (C) firms with potential overvaluation justifies the need for strategic asset allocation. The integration of factor modelling, such as the Fama-French approach, provides a structured method to assess and manage risk-adjusted returns while aligning with evolving ESG considerations. As the financial sector has increasing influence from regulatory and investor-driven sustainability trends, the retail mutual fund must employ comprehensive risk management strategies to navigate the complexities of pollution-related investment decisions while maintaining long-term profitability and environmental responsibility.

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